



Department of Computer Science

Lake Shore Campus

Doyle Hall

Syllabus – COMP 271 and COMP 400B Section 001 Data Structures I

Course Description

In this course, you will improve your programming skills, learn about object oriented design, and explore fundamental data structures such as arrays, linked lists, stacks, queues, trees, and hash tables. You will look at how these data structures are implemented and how to use them effectively to solve computational problems.

The course emphasizes algorithm analysis, efficiency, and practical applications using the Python programming language. You will also gain experience in testing and documenting code, preparing you for more advanced studies in computer science and software development.

Data Structures: Understand and implement fundamental data structures including arrays, linked lists, stacks, queues, trees, and hash tables. Analyze their performance characteristics and choose appropriate structures for different applications.

Object-Oriented Programming: Apply object-oriented programming principles such as encapsulation, inheritance, and polymorphism to design and implement modular and reusable code in Python.

Algorithm Analysis: Evaluate the time and space complexity of algorithms using Big-O notation. Analyze the efficiency of common algorithms for searching, sorting, and manipulating data structures.

Practical Programming Skills: Develop proficiency in Python programming, including testing and documenting code. Gain experience with version control systems and collaborative coding practices.

Problem Solving: Formulate and solve computational problems using appropriate data structures and algorithms. Develop critical thinking skills to approach complex programming challenges.

COMP 400B meets co-located with COMP 271 and is for use by graduate students doing prerequisites.

Prerequisites

- COMP 170, Introduction to Object Oriented Programming with a passing grade
- COMP 163, Discrete Structures with a passing grade

Course Schedule

The preliminary COMP 271 course schedule will be posted on Sakai in the same area as this Syllabus. Changes may occur based on class needs; any changes will be announced in class and updates posted in Sakai.

NOTE: The syllabus and course schedule are subject to change; changes will be announced in advance in class and posted on Sakai.

Special Course Requirements

1. COMP 271 is a programming intensive course. Considerable time will be spent creating programs in and outside class. A major portion of your performance in this course will be determined by your success in writing, reading, running, and testing these programs and how you explain and discuss programming with the class.
2. The course uses Sakai to organize materials. You will generally submit work using Sakai and receive feedback through Sakai. Ask for help if you are not familiar with Sakai.

Course Materials

Textbook / Required Reading: There is no required textbook for this course. All course materials, including lecture notes, assignments, and readings, are provided online through the course website and GitHub repository.

Programming environment GitHub Codespaces and Visual Studio Code are recommended programming environments for this course. Students are welcome to use any programming environment they prefer, provided it supports Python development and they are comfortable using it for course assignments and projects.

Course Components and Assessment

Course performance is evaluated with a series of assignments. Assignments are typically organized as homework, quizzes, and exams. In a 200-level course and higher, homeworks and exams are week-long, take-home assignments. Depending on the course, there are between 8 and 12 assignments per term.

Assessment approach (aligned with the coordinator's model): In this course, assessment emphasizes learning through feedback and iterative improvement. You will receive feedback about what you did well and what you need to work on; your adherence to feedback across the semester, as demonstrated in subsequent work, will be an important factor in determining your course grade. At midterm and/or near the end of the course, you may be asked to submit a brief reflection statement about your learning and progress.

Important: While Sakai may show points for submission tracking, these do not necessarily represent evaluative scores in the traditional sense. Details on how course performance is evaluated and what constitutes strong performance will be clarified on Sakai and in class.

Expectations and Timely Completion

You are expected to complete all assignments, readings, presentations, labs, and projects on time and with correct process for submitting them. You are required to be prepared to participate in classroom and online discussions on all topics up to the current point in the class schedule.

You are required to be in all class sessions for the full class time. If for any reason you do miss a class session, it is your responsibility to determine what you missed, locate any handouts, determine any changes in assignments, course plans, or schedules, etc. It is not my obligation to help you make up for missing class.

Deadlines and late work: Deadlines are posted on Sakai. Late work is generally not accepted. Bona fide emergencies are handled on a case-by-case basis.

Academic Honesty

Students are expected to have read the statement on academic integrity available in the Loyola University Chicago academic catalog. This policy applies to the course.

Academic dishonesty includes, but is not limited to, working together on assignments that are not group assignments, copying or sharing assignments or exam information with other students except in group assignments, submitting as your own information from current or former students of this course, copying information from anywhere on the web and submitting it as your own work, and

submitting anything as your own work which you have not personally created for this course. If you do wish to use materials that are not your own, please check with your instructor ahead of time and cite your source clearly. When in doubt, ask first.

Generative AI Usage (e.g., ChatGPT)

Gaining fluency in programming requires trial/error and sustained practice. This course aims to help students develop core problem solving skills, computational thinking, and programming fluency. Students are discouraged from using generative AI as a primary means of obtaining solutions.

If a student chooses to use generative AI to obtain *hints* or to clarify concepts, they must cite which tool was used and describe how the output was evaluated for correctness. Any such assignment may have its evaluation affected based on the extent to which AI was used to produce the work.

Use of generative AI is strictly forbidden on all course quizzes and exams.

Online Recording Policy

In this class, software may be used to record live class discussions. As a student in this class, your participation in live class discussions may be recorded. These recordings will be made available only to students enrolled in the class, to assist those who cannot attend the live session or to serve as a resource for those who would like to review content that was presented. All recordings will become unavailable to students in the class when the Sakai course is unpublished.

In cases where an online session is recorded, students who prefer to participate via audio only will be allowed to disable their video camera so only audio will be captured. Please discuss this option with your professor.

Religious Holidays

If you need to miss class for a religious holiday, please alert your instructor in advance so that it will not be counted as an absence. As needed, conflicting due dates for quizzes and tests, homework, and lab assignments can be adjusted with advanced notice.

Gender Misconduct Mandatory Reporting

Loyola University Chicago faculty are committed to supporting our students and upholding gender equity laws as outlined by Title IX. Therefore, if a student chooses to confide in a member of Loyola's faculty or staff regarding an issue of gender-based misconduct, that faculty or staff member

is obligated to notify the University's Title IX office, which will assist the student in connecting with resources.

Students with Disabilities

Loyola University provides reasonable accommodations for students with disabilities. Any student requesting accommodations related to a disability or other condition is required to register with the Student Accessibility Center (SAC), located in Sullivan Center, Suite 117. Students will provide professors with an accommodation notification from SAC, preferably within the first two weeks of class. Students are encouraged to meet with their professor individually in order to discuss accommodations. All information will remain confidential.

Students with Sponsorships and Scholarships

If you require a certain grade in order to satisfy a sponsor or a scholarship requirement, please be sure to monitor your course standing and course expectations as described in this syllabus and on Sakai. We will consider only your performance in this course. You are responsible to monitor your progress and to keep apprised of withdrawal dates posted by the registrar.

Office Hours and Help

Office hours by appointment via Calendly: <https://calendly.com/kennaoui-luc>, in Doyle Center Room 304 or via Zoom.

Extra time and special arrangements on request. You are encouraged to set up times to talk with the instructor and TA about the course, programming careers, and questions you have.

The Department of Computer Science provides tutors for this and other courses; you can find information about tutors on the CS website.

Continuous Improvement

The department believes in a quality process of continuous improvement. Anything can be improved by applying the quality process of "Plan, Do, Check, Act" (PDCA). To improve the course and the learning of programming and computer science concepts, your feedback, comments, suggestions, and complaints are welcome at any time.

In support of this PDCA process above, the instructor may ask you to participate in surveys and discussions during the course. These surveys will measure student impressions of the course; when

time permits, results may be shared with the class. Your inputs on these surveys are anonymous and in no way affect your course standing.

COMP 271 – Syllabus (Spring 2026)

Version 1.0, generated for Spring 2026 based on department syllabus structure.